

Data Analytics – Exercises

(Week 07)

In these exercises, you will learn:

- to statistically analyze contingency tables using the Chi-squared test
- to perform a Pearson correlation including a significance test

In the data analytics process model, these exercises cover part of the steps “Statistical data analysis and/or Modeling” and “Evaluation & Interpretation” (see figure 1). Results of the exercises must be uploaded as separate files (**no .zip files!**) by each student on Moodle. Details on how to submit the results can be found in the tasks below.

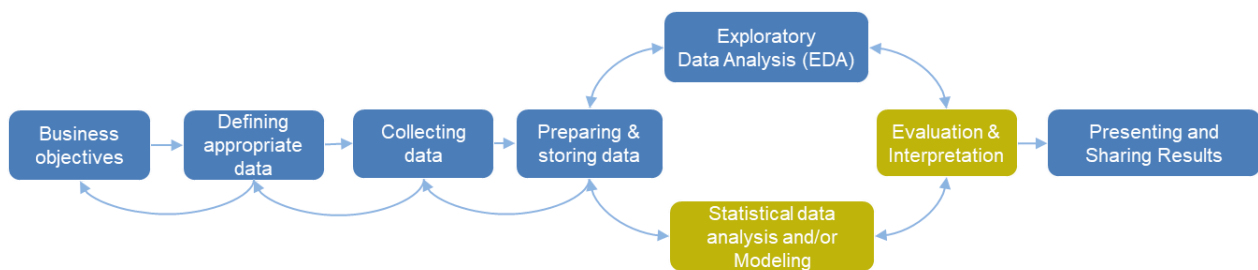


Figure 1: Data analytics process model (see slides of week 01)

Task 1

In this exercise, you will learn to statistically analyze a contingency table using the Chi-squared test. The tasks are:

- Run the Jupyter notebook '[analysis_of_contingency_tables.ipynb](#)' and try to find out, what the Python code does.
- Look at the Chi-squared test for the contingency table showing the number of apartments for different living area and price categories. The Chi-Square Test is commonly used to test statistical association between two categorical variables.
- Do you fully understand, the output and interpretation of the results of the Chi-squared test? If not, look at the slides provided in the large class of this week.

Task 2

In this exercise, you will learn to calculate a Chi-squared test “by hand”.

- a) Follow this [link](#) to a research study in which a Chi-squared has been used to study animal behavior. In detail, the research is based on the social interactions of two spotted hyenas (*Crocuta crocuta*) which were observed over 40 hours in a U.S. zoo. The 4x2 contingency table below has been taken from this research study.

	Female	Male	Total
Fear	0	4	4
Greeting	7	6	13
Aggression	9	0	9
No Response	15	10	25
Total	31	20	51

- b) In the Jupyter notebook of task 1, you will find a section ‘Applying the Chi-squared test to animal behavior’. Go to this section and perform a Chi-squared test based on the table above (note that the table is already included in the Jupyter notebook).
- c) Write a Python function which calculates the expected frequencies of the contingency table. Input must be a numpy array or data frame with observed frequencies. To write the function, ask ChatGPT for help. Apply the function to your data.
- d) Compare your calculated expected frequencies with those from the `chi2_contingency()` method in Python.
- e) Extend the Python function under c) to additionally provide the Chi-square test statistic. Again, ask ChatGPT for help.
- f) Compare your calculated Chi-square test statistic with the Chi-square test statistic from the `chi2_contingency()` method in Python.
- g) In the Jupyter notebook, state in one sentence whether the results of the original study are correct or not.

Task 3

In this exercise, you will learn to perform correlation analyses. The tasks are:

- a) Run the Jupyter notebook '[correlation_analysis.ipynb](#)' and try to find out, what the Python code does.
- b) In the Jupyter notebook section 'Correlation analysis based on car data', part of the car data set '[autoscout24_data.csv](#)' has already been imported to a data frame.
- c) Explore the numerical car data (i.e.: Price, Kilometer, PS) exploratively using a paired scatterplot (see slides and exercises of previous weeks).
- d) Perform correlation analyses inclusive significance tests for the variables:
 - Price versus Kilometer
 - Price versus PS
 - Kilometer versus PS